Experiment 1: Develop a program to blink 5 LEDs back and forth.

Pin Connections:

LED	ESP32 Pin	Arduino UNO Pin
L1	GPIO 2	D5
L2	GPIO 4	D6
L3	GPIO 5	D7
L4	GPIO 18	D8
L5	GPIO 19	D9

Code:

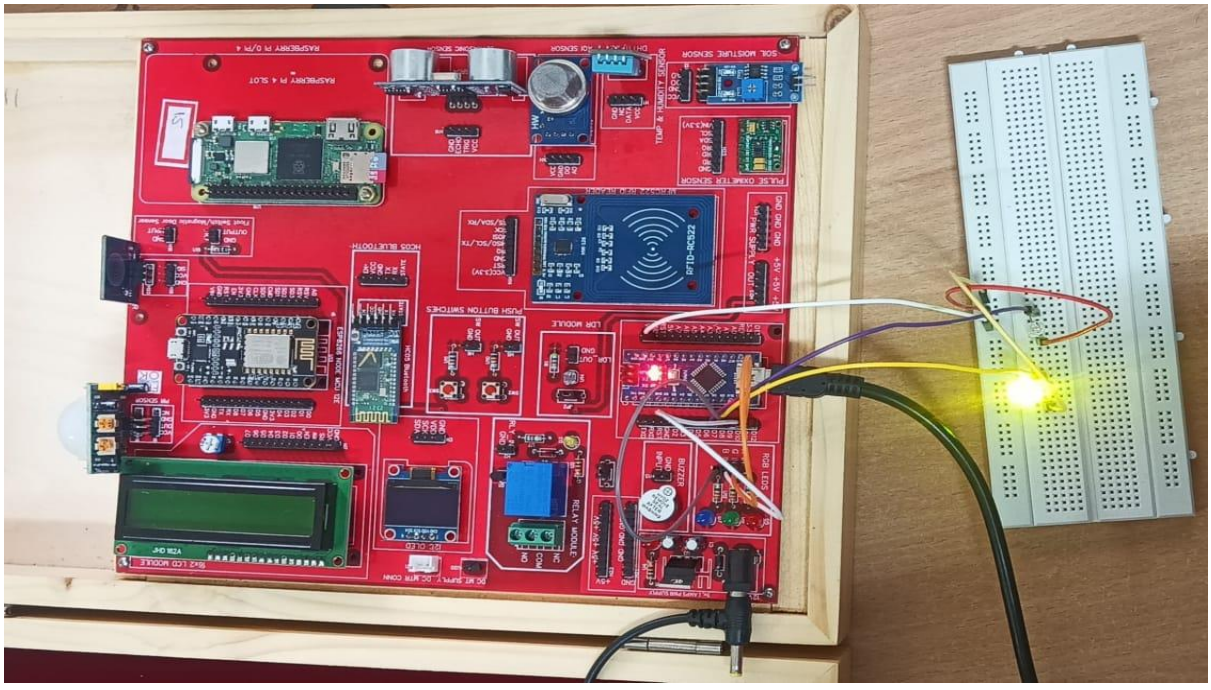
```
void setup()
 {
  pinMode(5,OUTPUT);
  pinMode(6,OUTPUT);
  pinMode(7,OUTPUT);
  pinMode(8,OUTPUT);
   pinMode(9,OUTPUT);
}
void loop()
 {
  digitalWrite(5,HIGH);
  delay(1000);
  digitalWrite(5,LOW);
  delay(1000);
  digitalWrite(6,HIGH);
  delay(1000);
  digitalWrite(6,LOW);
  delay(1000);
```

```
  digitalWrite(7,HIGH);
  delay(1000);
  digitalWrite(7,LOW);
  delay(1000);

 digitalWrite(8,HIGH);
  delay(1000);
  digitalWrite(8,LOW);
  delay(1000);

  digitalWrite(9,HIGH);
  delay(1000);
  digitalWrite(9,LOW);
  delay(1000);
}
```

Circuit Diagram:



Experiment 2: Develop a program to interface a relay with Arduino board.

Pin Connections:

Relay	ESP32 Pin	Arduino UNO Pin
Relay 1/2	GPIO 15	D7

Code:

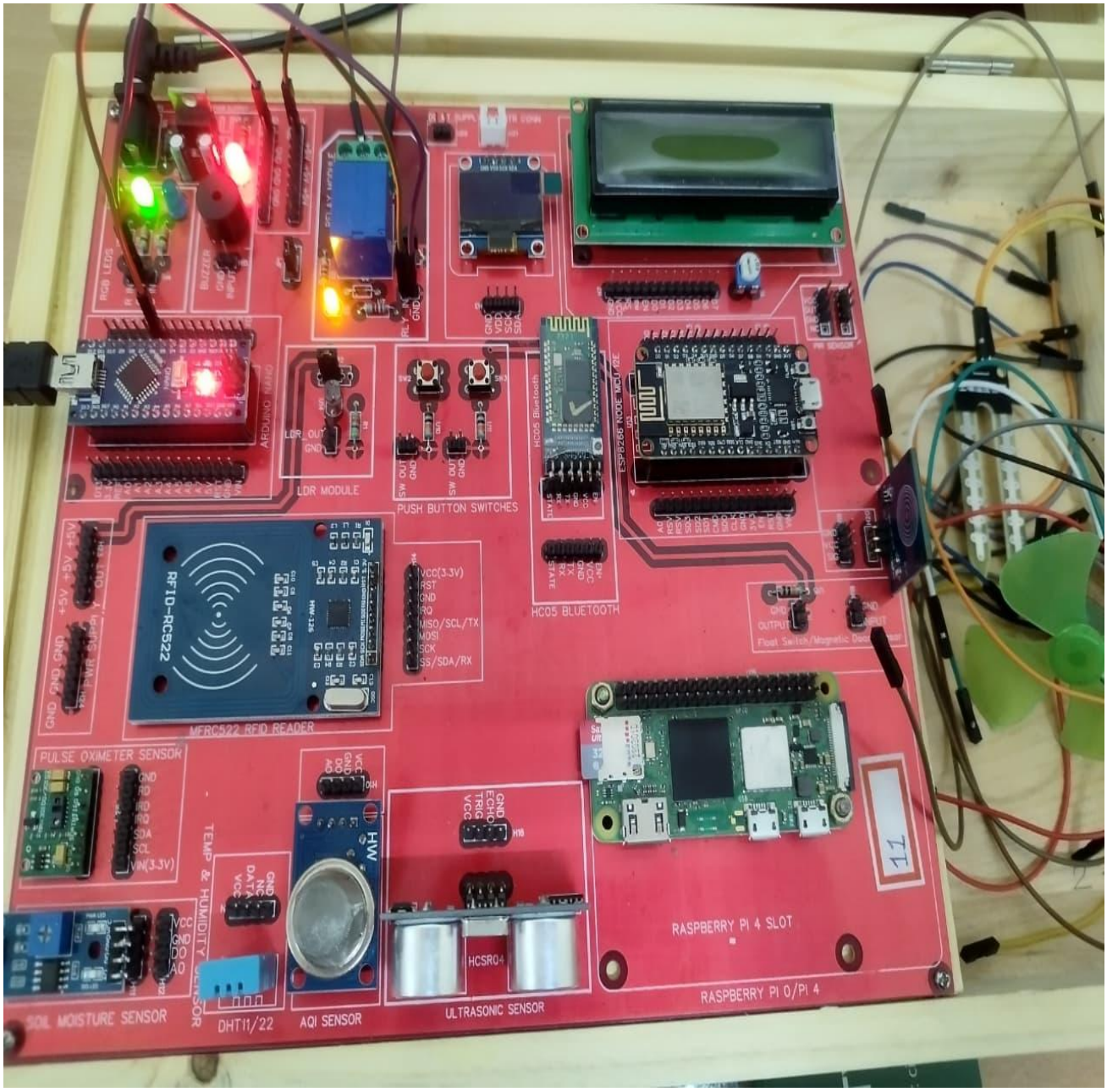
```
// Define the relay control pin
const int relayPin = 7;  // You can use any digital pin


void setup() {
  pinMode(relayPin, OUTPUT);      // Set the relay pin as output
  digitalWrite(relayPin, HIGH);   // Turn relay   On
}


void loop() {
  digitalWrite(relayPin, LOW);    // Turn    OFF the relay
  delay(1000);                    // Wait for 2 seconds


  digitalWrite(relayPin, HIGH);   // Turn    Relay On
  delay(1000);                    // Wait for 2 seconds
}
```

Circuit:


LDR MODULE
PUSH BUTTON SWITCHES
HC05 BLUETOOTH
RFID-RC522
MFRC522 RFID READER
PULSE OXIMETER SENSOR
TEMP & HUMIDITY SENSOR
SOIL MOISTURE SENSOR
DHT11/22
AQI SENSOR
HCSR04
ULTRASONIC SENSOR
RASPBERRY PI 4 SLOT
RASPBERRY PI 0/PI 4

Experiment 3: Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensor.

Pin Connections:

Component		ESP32 Pin	Arduino UNO Pin
Ultrasonic	TRIG	GPIO 13	D9
	ECHO	GPIO 12	D10
Sound Sensor	OUTPUT Pin	GPIO 14	D8
Buzzer	Buzzer pin	GPIO 15	D7

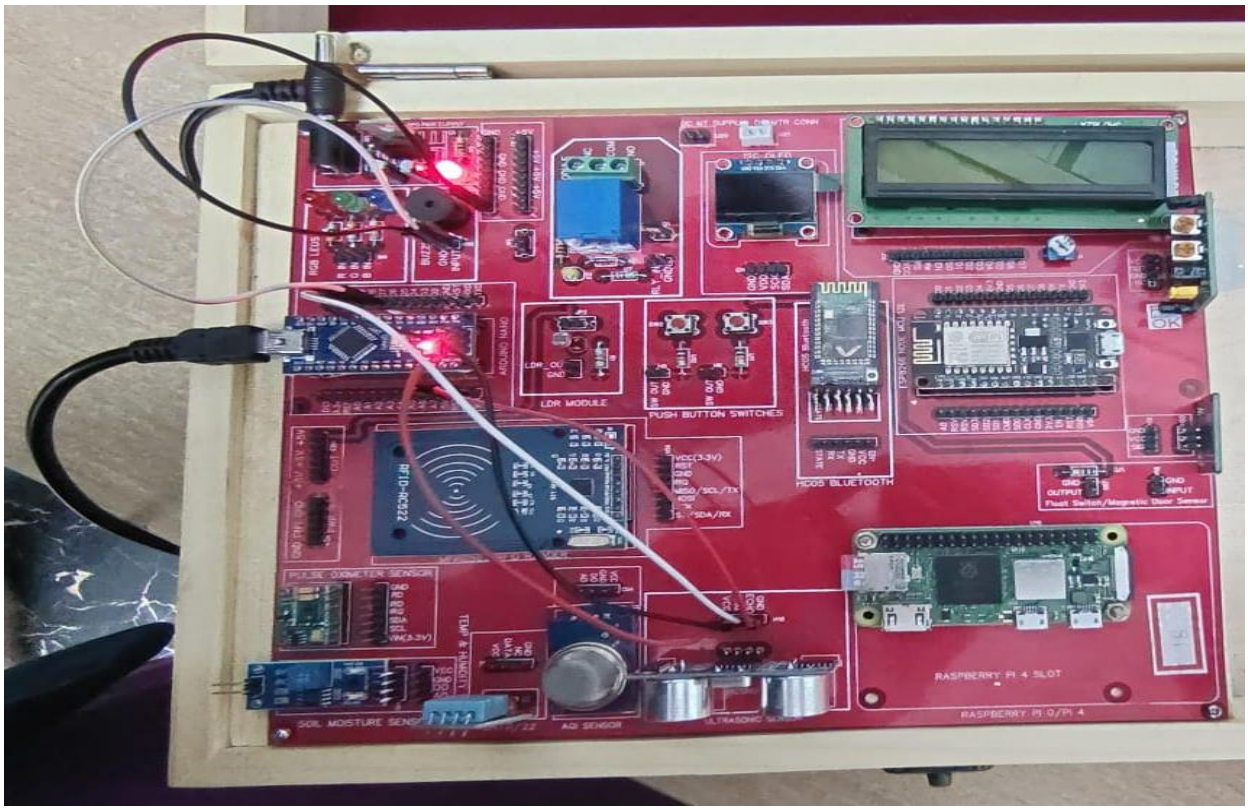
Code:

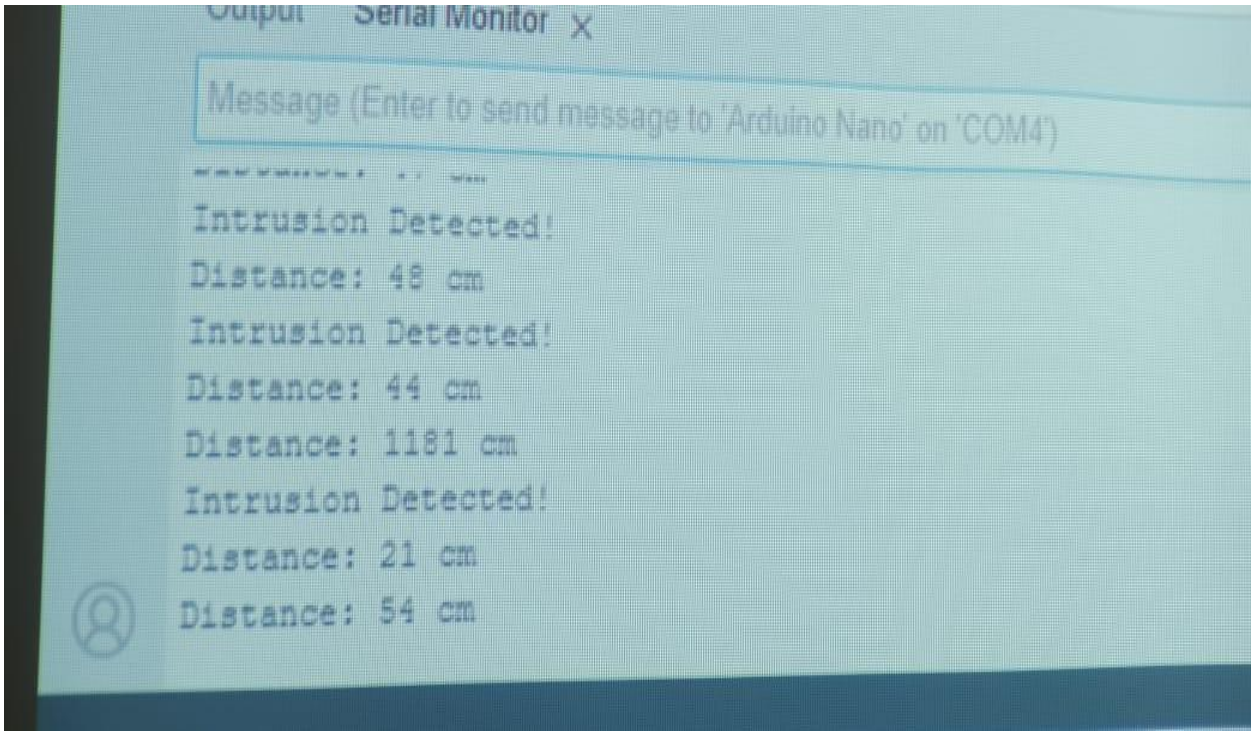
```
//Pin definitions
 const int trigPin=9;
 const int echoPin=10;
 const int buzzerPin=7;
 long duration;
 int distance;

 void setup()
{
 pinMode(trigPin,OUTPUT);
 pinMode(echoPin,INPUT);
 pinMode(buzzerPin,OUTPUT);
 Serial.begin(9600);
 }
```

```
 void loop()
{
//---Ultra sonic sensor distancemeasurement--
 digitalWrite(trigPin,LOW);
 delayMicroseconds(2);
 digitalWrite(trigPin,HIGH);
 delayMicroseconds(10);
 digitalWrite(trigPin,LOW);
 duration=pulseIn(echoPin,HIGH);
 distance=duration*0.034/2;//Converttocm
//---IntrusionDetectionCondition--
 if(distance<50){//Threshold:50cmorsounddetected
     digitalWrite(buzzerPin,HIGH);//AlarmON
     Serial.println("IntrusionDetected!");
  }
  else
     {
       digitalWrite(buzzerPin,LOW); //AlarmOFF
     }
//Print distance for debugging
Serial.print("Distance:");
Serial.print(distance);
Serial.println("cm");
delay(500);
}
```

Circuit Diagram and Serial Monitor Output:





Experiment 4: Develop a program to control a DC motor with Arduino board.

Relay	ESP32 Pin	Arduino UNO Pin
Relay 1/2	GPIO 15	D7
Push Button 1/2	GPIO 4	D2

Motor to Relay:

- One terminal of the motor connects to **+ of power supply.**
- The other motor terminal connects to **COM (Common)** of the relay
- **NO (Normally Open)** of the relay goes to **GND** of power supply

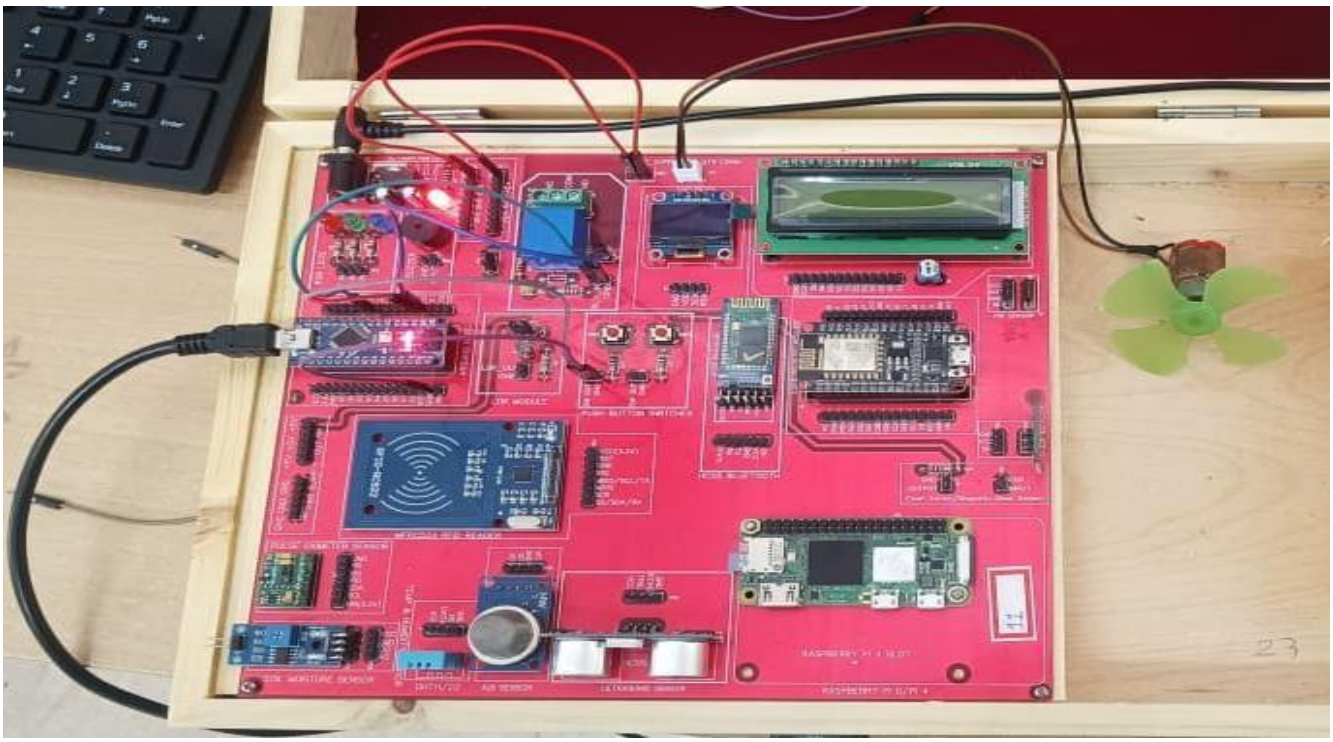
Code :

```
const int relayPin=7; //Relay control pin
 const int buttonPin=2; //Push button pin

 void setup()
 {
 pinMode(relayPin,OUTPUT);
 pinMode(buttonPin,INPUT);
 Serial.begin(9600);
 }

 void loop()
 {
 int buttonState=digitalRead(buttonPin);
 Serial.println(buttonState);
   if(buttonState==1)
   {
     digitalWrite(relayPin,HIGH);
   }
 else
   {
     digitalWrite(relayPin,LOW);
   }
 }
```

Circuit Diagram:



Serial Monitor Output:



Experiment 5: Develop a program to deploy smart street light system using LDR sensor.

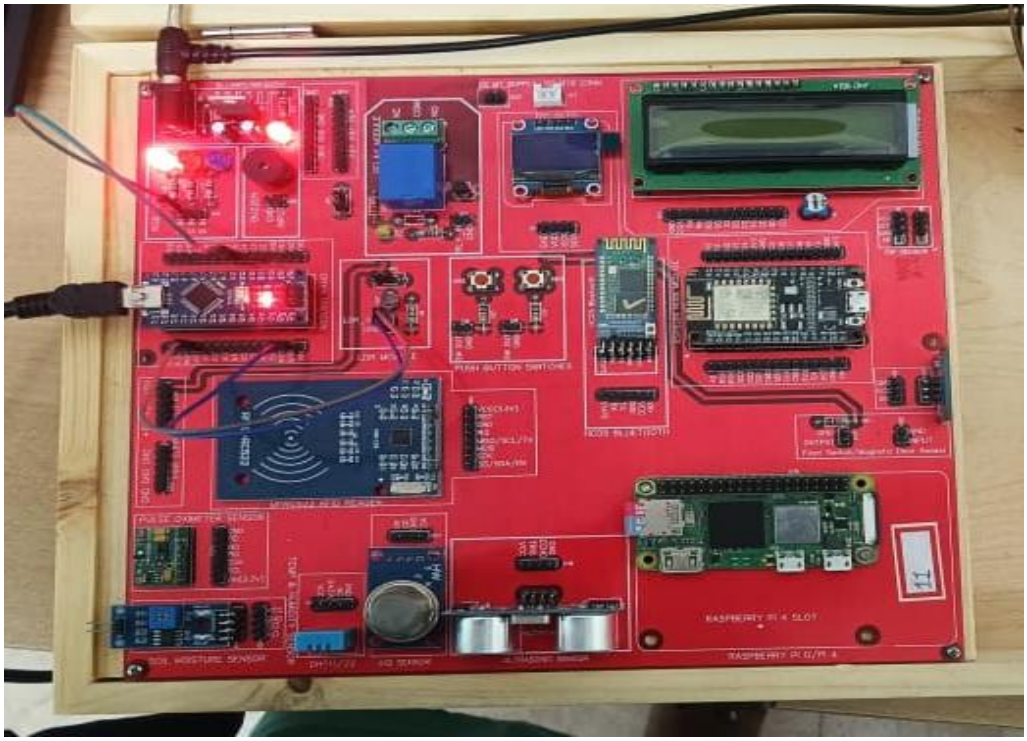
Pin Connections:

Component	ESP32 Pin	Arduino UNO Pin
LED	GPIO 25	D5
LDR	GPIO 34	A0

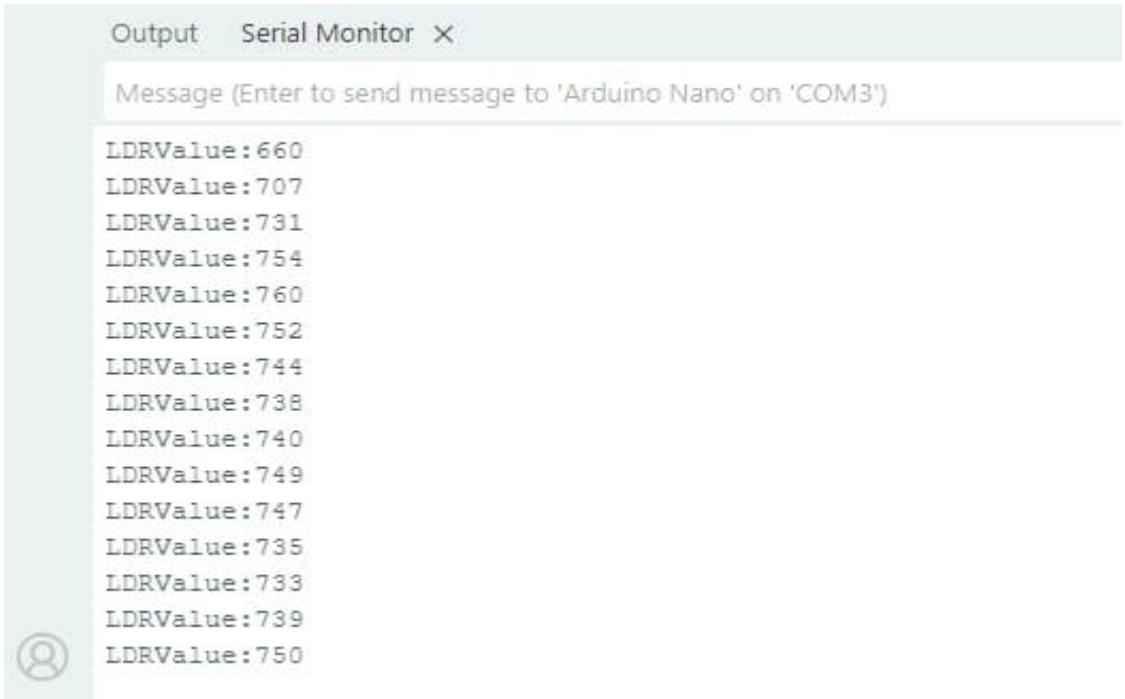
Code:

```
const int ldrPin=A0; //LDR connected to analog pin A0
const int lightPin=5; //LED relay control pin
int lightThreshold=1000; //Adjust threshold based on environment
void setup(){
pinMode(ldrPin,INPUT);
pinMode(lightPin,OUTPUT);
Serial.begin(9600);
}
void loop()
{
int ldrValue=analogRead(ldrPin);
Serial.print("LDRValue:");
Serial.println(ldrValue);
if(ldrValue<lightThreshold){
digitalWrite(lightPin,HIGH);//Turn ON light(dark environment)
}
else
{
digitalWrite(lightPin,LOW); //Turn OFF light(bright environment)
}
delay(500);
}
```

Circuit Diagram:



Serial Monitor Output:



Experiment 6: Develop a program to classify dry and wet waste with the moisture sensor.

Pin Connections:

Component	ESP32 Pin	Arduino UNO Pin
Moisture Sensor A0	GPIO 32	A0

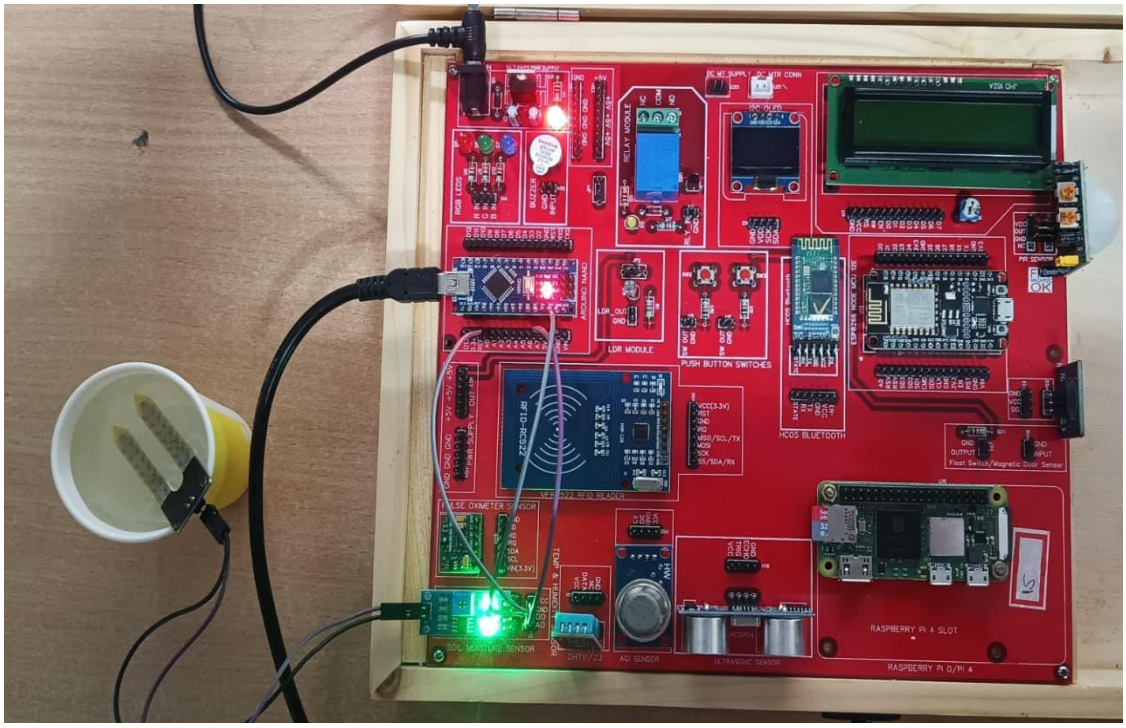
Code :

```
#define MOISTURE_SENSOR_PIN A0 //Use A0 for ArduinoUno

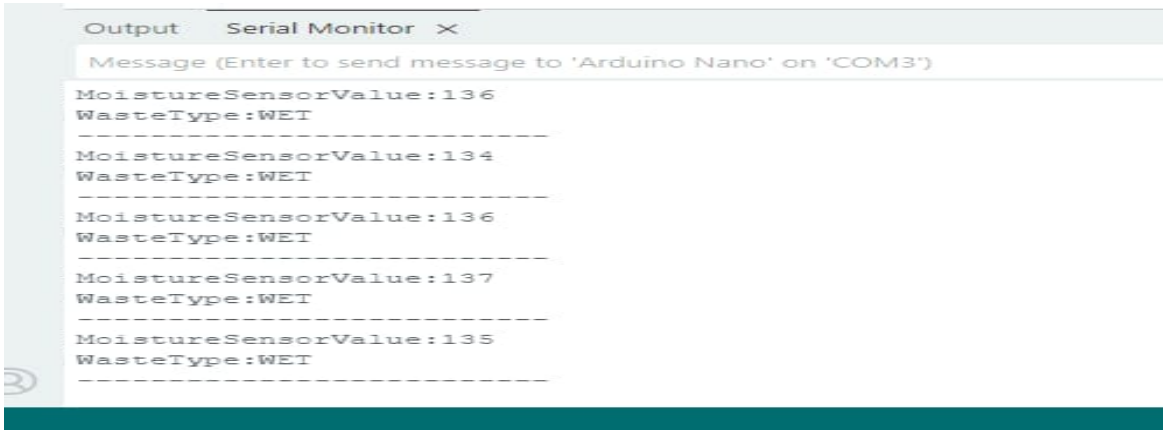

 void setup(){
 Serial.begin(115200);
 delay(1000);
 Serial.println("Moisture-BasedWasteClassificationStarted...");
 }

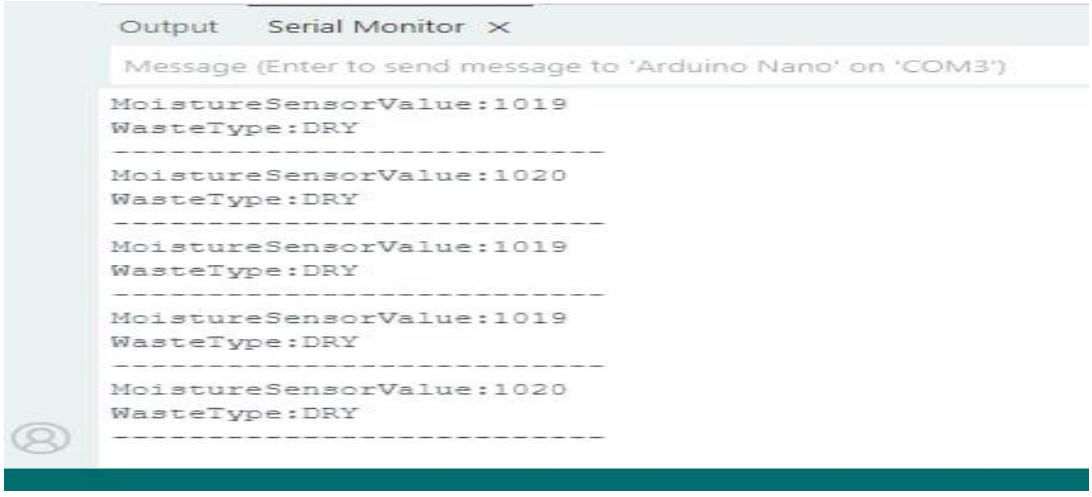

 void loop(){
 int moistureValue=analogRead(MOISTURE_SENSOR_PIN);
 Serial.print("MoistureSensorValue:");
 Serial.println(moistureValue);
 if(moistureValue>400){
 Serial.println("WasteType:DRY");
 }else{
 Serial.println("WasteType:WET");
 }
 Serial.println("----------------------");
 delay(2000);//Delay for readability
 }
```

Circuit Diagram:



Serial Monitor Output:





7. Develop a program to read the PH value of a various substances like milk, lime and water.

Pin Connections:

Component	ESP32 Pin	Arduino UNO Pin
+ 5V	GPIO 34	+ 5V
+ O/P	5V	A0
GND	GND	GND

Code :

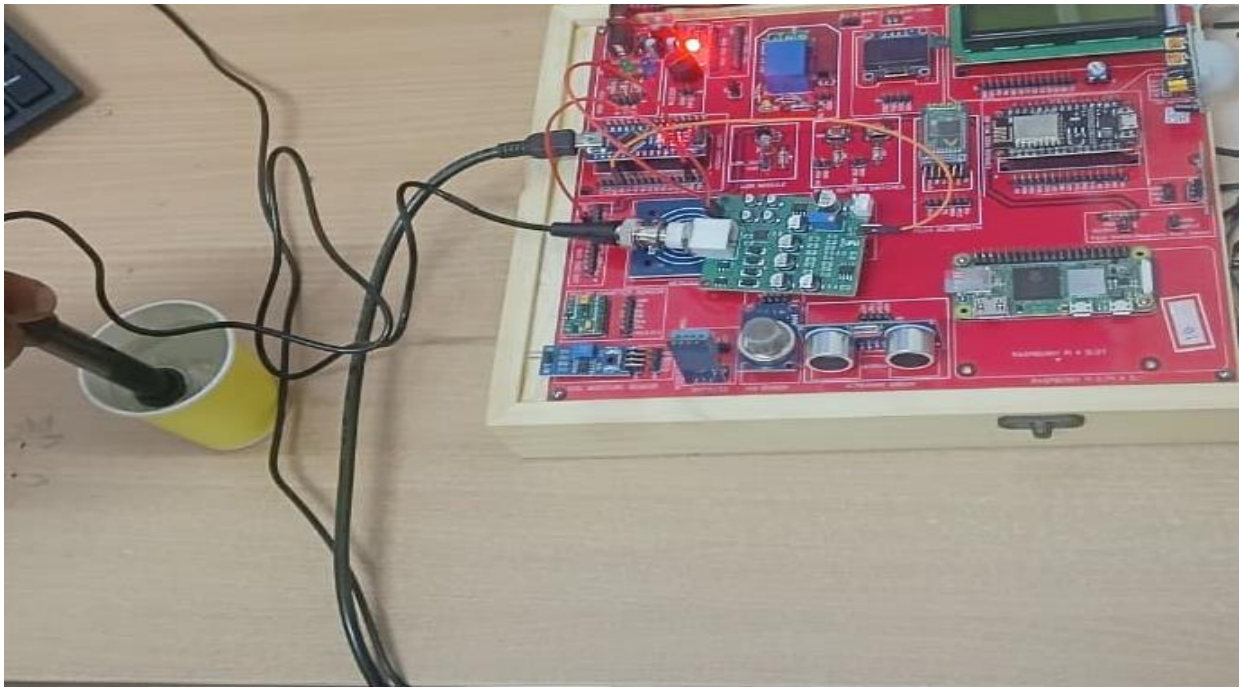
```
#define PH_PIN A0   // Analog pin for pH sensor

void setup() {
  Serial.begin(9600);
}

void loop() {
  int rawValue = analogRead(PH_PIN);         // Read analog value (0-1023)
  float voltage = rawValue * (5.0 / 1023.0); // Convert to voltage (0-5V)

  // Convert voltage to pH value
  // The formula may vary depending on your sensor
  float pH = 7 + ((2.0 - voltage) / 0.18); // Basic formula (approximate)
  Serial.print("Raw ADC: ");
  Serial.print(rawValue);
  Serial.print("  Voltage: ");
  Serial.print(voltage, 2);
  Serial.print(" V  =>  pH: ");
  Serial.println(pH, 2);
  delay(1000); // Read every second
}
```

Circuit Diagram:



Serial Monitor Output:

```
Raw ADC: 426  Voltage: 2.08 V  =>  pH: 6.54
Raw ADC: 432  Voltage: 2.11 V  =>  pH: 6.38
Raw ADC: 439  Voltage: 2.15 V  =>  pH: 6.19
Raw ADC: 430  Voltage: 2.10 V  =>  pH: 6.44
Raw ADC: 421  Voltage: 2.06 V  =>  pH: 6.68
Raw ADC: 421  Voltage: 2.06 V  =>  pH: 6.68
Raw ADC: 424  Voltage: 2.07 V  =>  pH: 6.60
Raw ADC: 433  Voltage: 2.12 V  =>  pH: 6.35
```

Experiment 8: Develop a program to detect the gas leakage in the surrounding environment

Pin Connections:

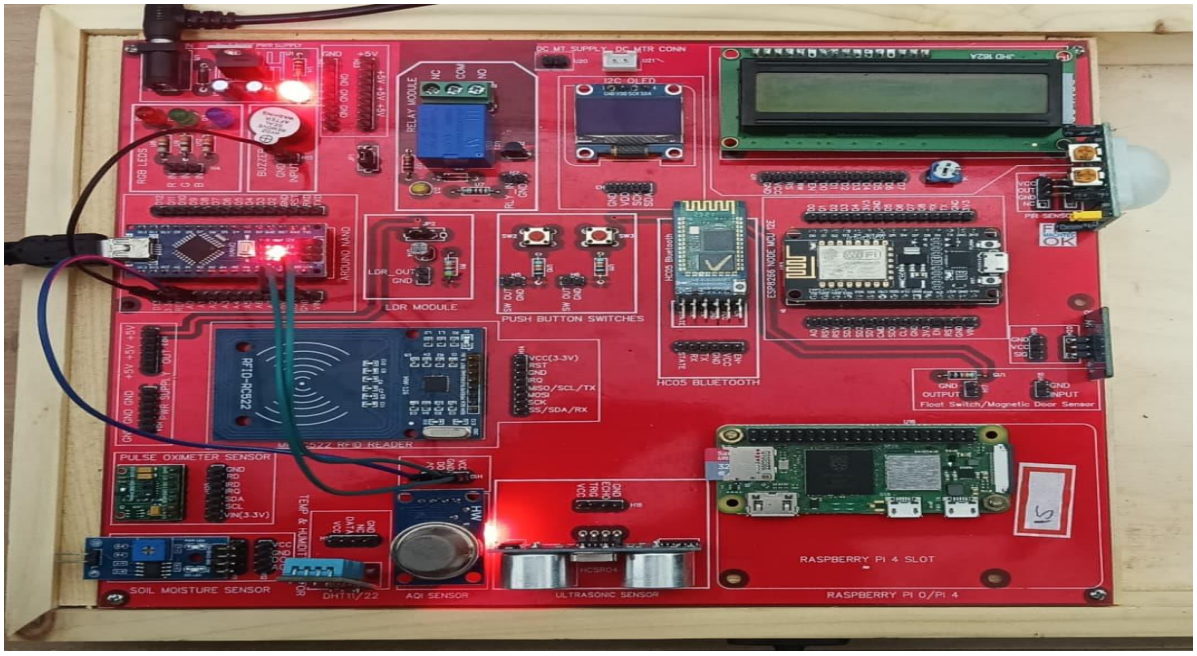
Component	ESP32 Pin	Arduino UNO Pin
MQ135 A0	GPIO 15	A0
Buzzer	GPIO 4	D13

Code:

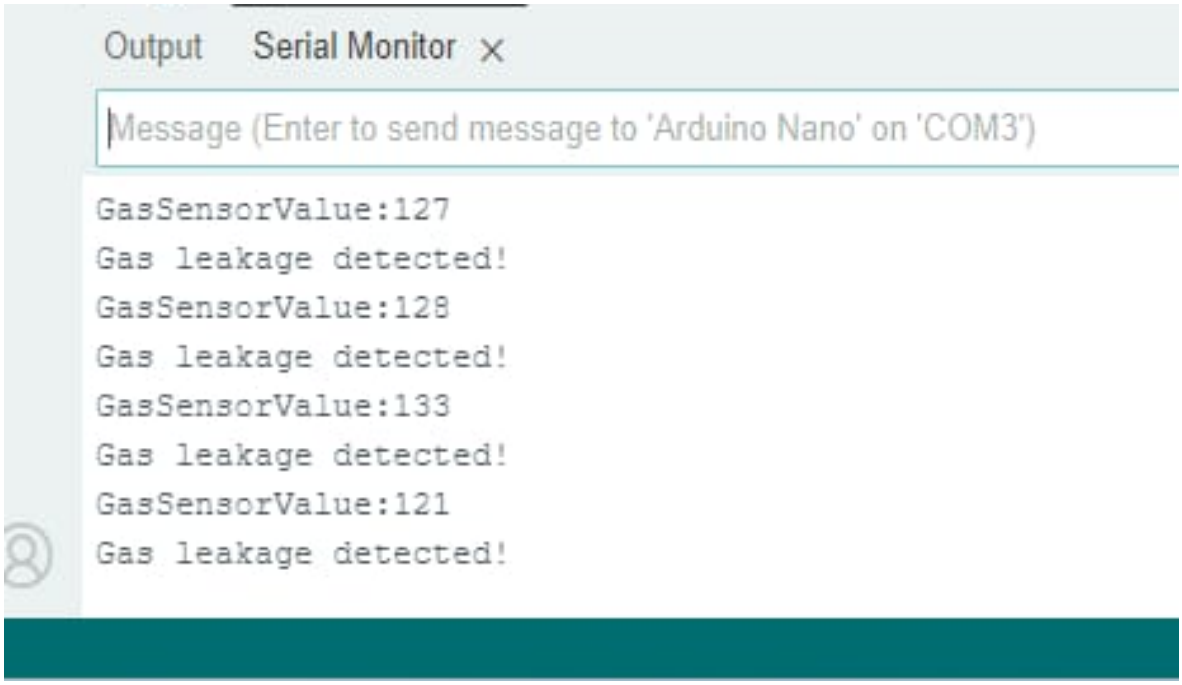
```
const int gasSensorPin=A0; //Use GPIO34 or GPIO36 if using ESP32
 const int buzzerPin=13; //Connect buzzer to digitalpin


 void setup(){
 Serial.begin(115200);
 pinMode(buzzerPin,OUTPUT);
 }
 void loop(){
 int gasValue=analogRead(gasSensorPin);
 Serial.print("GasSensorValue:");
 Serial.println(gasValue);
 if(gasValue>100){//Threshold may vary depending on environment
 digitalWrite(buzzerPin,HIGH);
 Serial.println("Gas leakage detected!");
 }else{
 digitalWrite(buzzerPin,LOW);
 }
 delay(1000);
 }
```

Circuit Diagram:



Serial Monitor Output :



Experiment 9: Develop a program to demonstrate weather station readings using Arduino.

Pin Connections:

Component	ESP32 Pin	Arduino UNO Pin
DHT 11 O/P	GPIO 4	D4

Code :

```
#include<DHT.h>
#include<Wire.h>

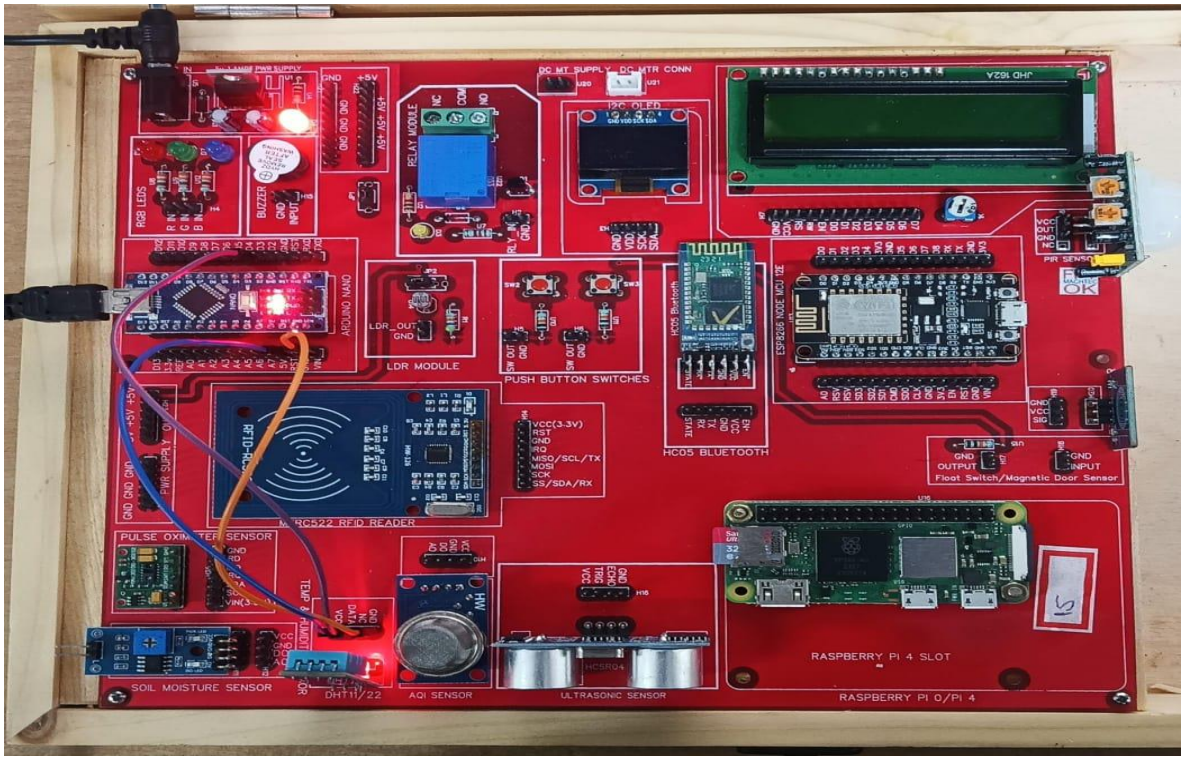
 #define DHTPIN 4
 #define DHTTYPE DHT11
 DHT dht11(DHTPIN,DHTTYPE);

 void setup(){
 dht11.begin();
 Serial.begin(9600);

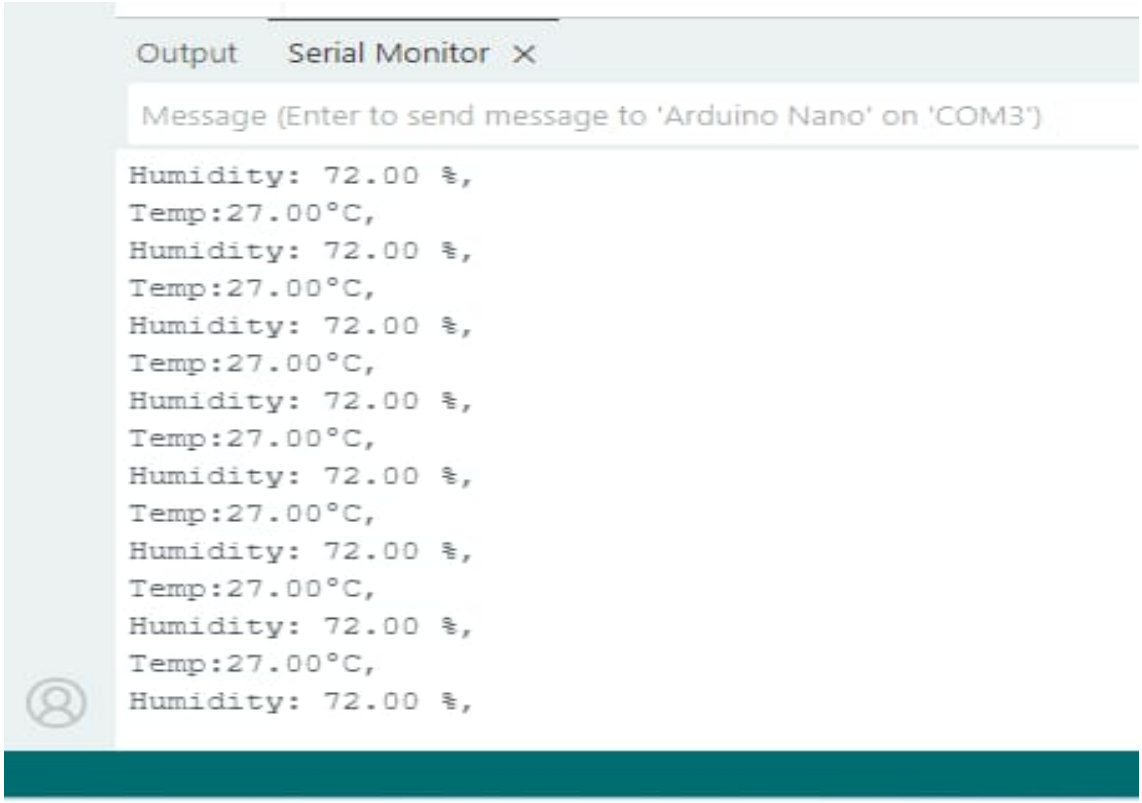
  }
 void loop(){
 float temp=dht11.readTemperature();
float hum=dht11.readHumidity();

 Serial.print("Temp:");Serial.print(temp);Serial.println("°C,");
 Serial.print("Humidity: "); Serial.print(hum); Serial.println(" %, ");
 delay(500);
 }
```

Circuit Diagram:



Serial Monitor Output:



Experiment 10: Develop a program to setup a UART protocol and pass a string through the protocol.

Pin Connections:

Component	ESP32 Pin	Arduino UNO Pin
LED	GPIO 2	D13

Code:

```
#define LED_PIN 2 // GPIO2 for ESP32, use 13 for Arduino Uno
String inputString = ""; // To store incoming serial data
bool stringComplete = false;

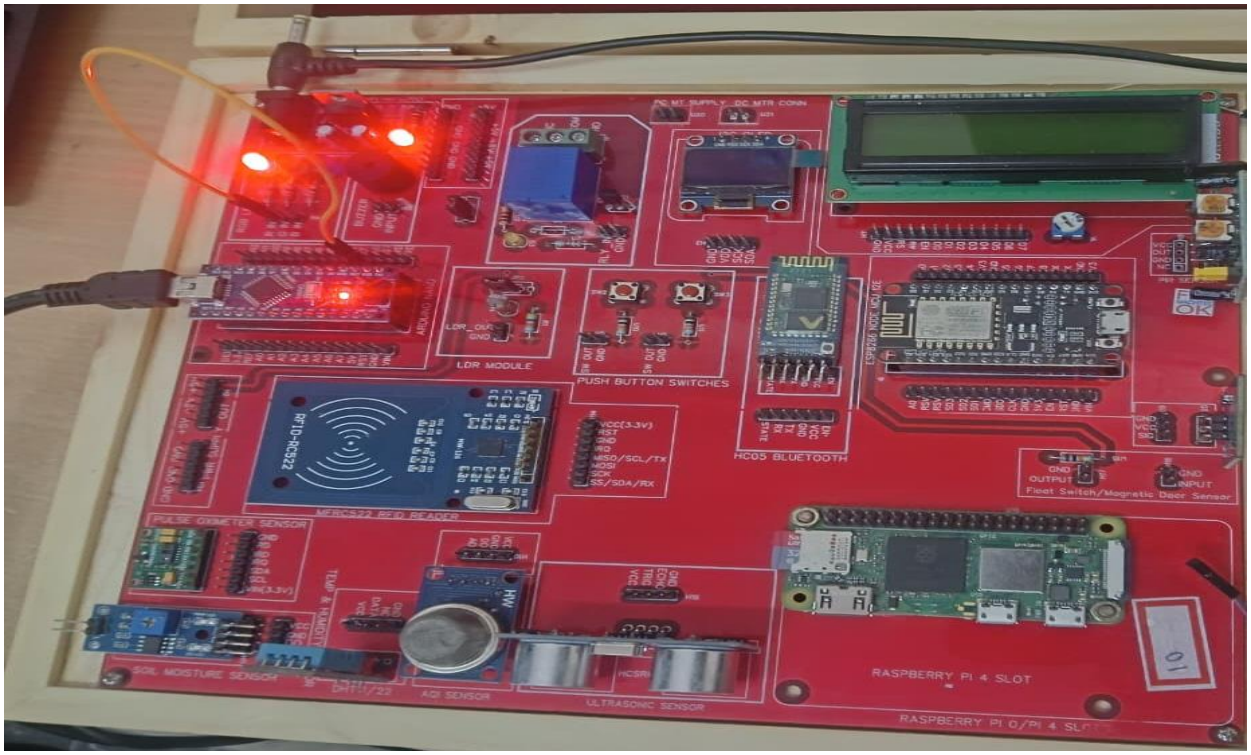
void setup() {
pinMode(LED_PIN, OUTPUT);
Serial.begin(9600);
Serial.println("UART Command Processor Initialized.");
inputString.reserve(200); // Optional: reserve memory for speed
}
void loop() {
// When a complete command is received
if (stringComplete) {
processCommand(inputString);
inputString = "";
stringComplete = false;
}
}

// Serial Event triggered by new data
void serialEvent() {
while (Serial.available()) {
char inChar = (char)Serial.read();
inputString += inChar;
```

```
// Check for end of line
if (inChar == '\n') {
stringComplete = true;
}
}
}

void processCommand(String command) {
command.trim(); // Remove newline or extra whitespace
if (command.equalsIgnoreCase("LED ON")) {
digitalWrite(LED_PIN, HIGH);
Serial.println("LED is turned ON");
}
else if (command.equalsIgnoreCase("LED OFF")) {
digitalWrite(LED_PIN, LOW);
Serial.println("LED is turned OFF");
}
else if (command.equalsIgnoreCase("STATUS")) {
int state = digitalRead(LED_PIN);
if (state == HIGH) {
Serial.println("LED STATUS: ON");
} else {
Serial.println("LED STATUS: OFF");
}
}
else {
Serial.print("Unknown Command: ");
Serial.println(command);
}
}
```

Circuit Diagram:



Serial Monitor Output:



Experiment 11: Develop a program to a water level detection system using Ultrasonic sensor.

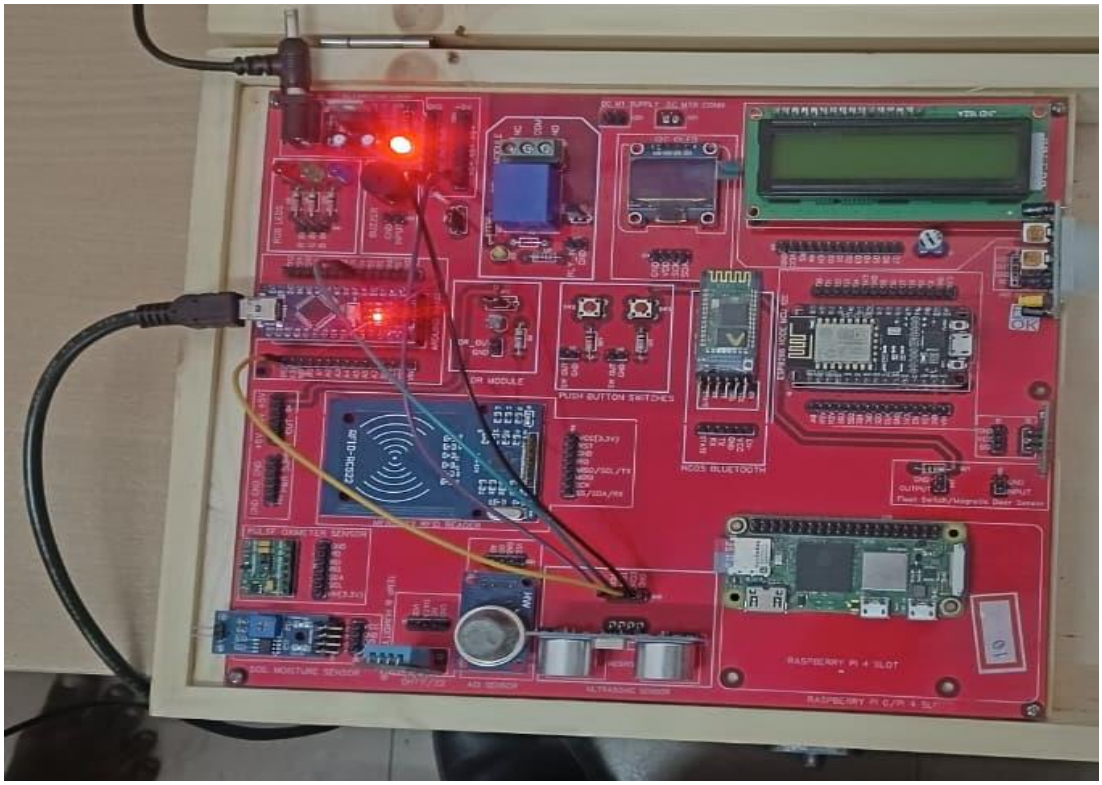
Pin Connections:

Component		ESP32 Pin	Arduino UNO Pin
Ultrasonic	TRIG	GPIO 5	D9
	ECHO	GPIO 18	D10

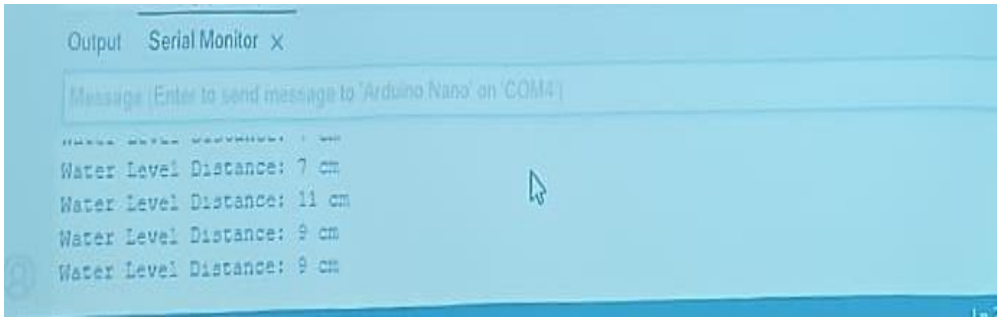
Code :

```
const int trigPin = 5;
const int echoPin = 13;
void setup() {
Serial.begin(115200);
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);
}
long getDistance() {
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);
long duration = pulseIn(echoPin, HIGH);
long distance = duration * 0.034 / 2;
return distance;
}
void loop() {
long distance = getDistance();
Serial.print("Water Level Distance: ");
Serial.print(distance);
Serial.println(" cm");
delay(1000);
}
```

Circuit Diagram:



Serial Monitor Output:



Experiment 12: Develop a program to simulate interfacing with the keypad module to record the keystrokes.

Pin Connections:

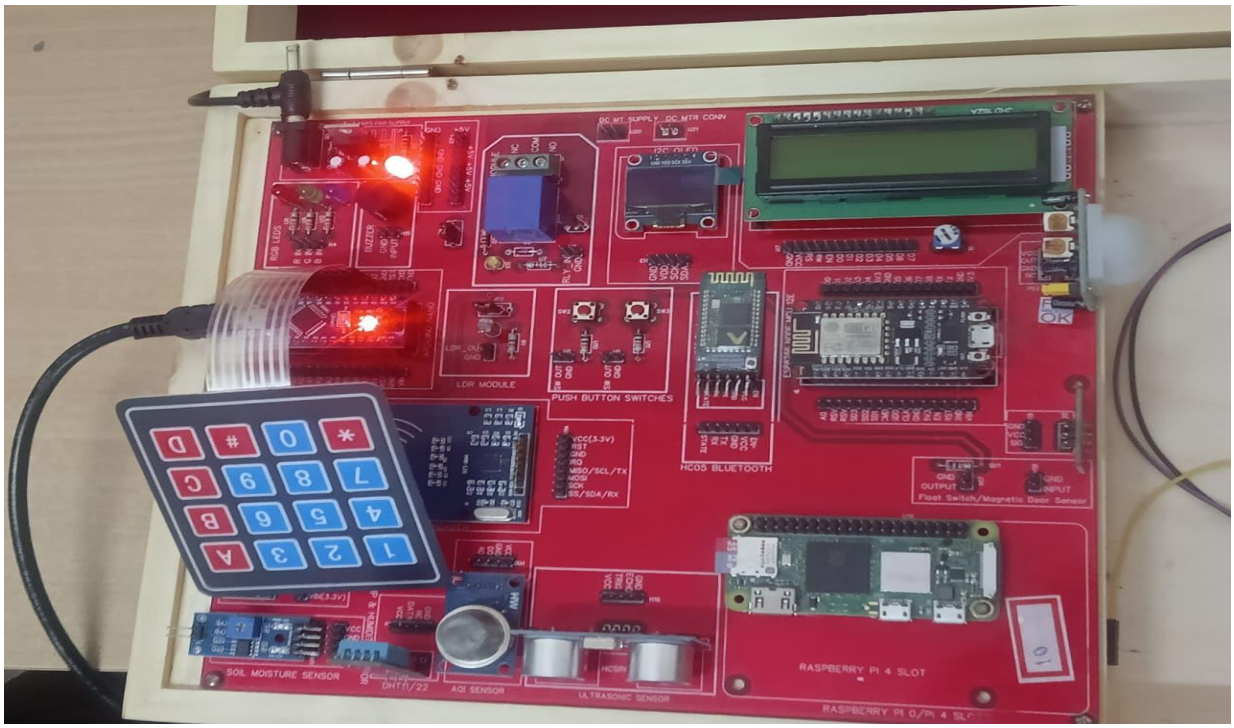
Component		ESP32 Pin	Arduino UNO Pin
Keypad	R1	GPIO 14	D2
	R2	GPIO 27	D3
	R3	GPIO 26	D4
	R4	GPIO 25	D5
	C1	GPIO 33	D6
	C2	GPIO 32	D7
	C3	GPIO 19	D8
	C4	GPIO 18	D9

Code:

```
#include <Keypad.h>
const byte ROWS = 4;
const byte COLS = 4;
char keys[ROWS][COLS] = {
{'1','2','3','A'}, {'4','5','6','B'}, {'7','8','9','C'}, {'*','0','#','D'}
};
byte rowPins[ROWS] = {2, 3, 4, 5}; // Modify as per your wiring
byte colPins[COLS] = {6, 7, 8, 9}; // Avoid GPIOs 34–39
Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);
void setup() {
Serial.begin(115200);
Serial.println("Keypad Ready");
}
```

```
void loop() {
char key = keypad.getKey();
if (key) {
Serial.print("Key Pressed: ");
Serial.println(key);
}
}
```

Circuit Diagram:



Serial Monitor Output:

